Evidence package synthesis

Build an AI evidence package from prior workshop results

Applied AI Design Workshop for Faculty
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Purpose: convert prior work into AI-ready evidence

What participants already created

- Problem statements
- Regression or classification metrics
- Baseline comparisons
- Data-preparation decisions
- Feature-engineering rationale
- Cluster profiles or PCA views

What AI needs to stay grounded

- Clear source context
- Metrics and visuals
- Known limitations
- Definitions of inputs and targets
- Boundaries on claims
- Human review criteria

Activity: purpose and outputs

Convert prior ML work into a concise evidence package that an AI system can use without inventing claims.

Day 3–4 evidence → Model result → Limitations → Grounded prompt → AI response

Completed package should answer:

- What evidence do we have?
- What does it support?
- What does it not support?
- How should an AI system use it?

Evidence package template

Problem

Dataset

Model or method

Metric/result

Visualization/table

Limitations

Human decision

Keep this packet concise: the AI system should use it as grounding context, not as a replacement for disciplinary judgment.

Guided scenario: predicting defective process runs

A faculty research group wants to determine whether process measurements can help identify manufacturing runs likely to produce a defect.

Element	Guided example
Intended user	Process engineer or laboratory researcher
Decision	Flag runs for additional inspection
Unit	One manufacturing process run
Target	Defect vs. acceptable
Model role	Decision support, not automatic approval

Key framing question: what decision would change if the evidence were reliable?

Step 1: state the problem precisely

Can process temperature, pressure, speed, and vibration be used to predict whether a manufacturing run will produce a defective part?

Field	Entry
Inputs	Temperature, pressure, speed, vibration
Target	0 = acceptable; 1 = defective
Task	Binary classification
Metric priority	Recall if missed defects are costly

Step 2: describe the dataset and class balance

Total process runs
220
rows / observations

Acceptable runs
176
80% of dataset

Defective runs
44
20% of dataset

Initial concern
Imbalance
accuracy may mislead

Variable	Meaning
temperature_c	Process temperature in degrees Celsius
pressure_kpa	Process pressure in kilopascals
speed_mm_s	Process speed in millimeters per second
vibration_mm_s	Measured vibration in millimeters per second
defect	Binary quality outcome

A majority-class model can appear accurate while detecting no defects.

Step 3: establish the baseline

Baseline rule: predict every process run as acceptable.

Baseline accuracy
0.800
correct mostly because defects are rare

Baseline defect recall
0.000
misses every defect

Interpretation

- Accuracy alone is not adequate for this problem.
- The baseline clarifies the minimum value a model must exceed.
- Defect recall is essential if missed defects are expensive.

Step 4: summarize the trained model

Model: standardized logistic-regression classifier

Data split: 75% training / 25% test · stratified · fixed random state

Metric	Result
Accuracy	0.855
Precision	1.000
Recall	0.273
F1 score	0.429

	Predicted acceptable	Predicted defect
Actual acceptable	44	0
Actual defect	8	3

The model produces few false alarms, but misses most actual defects.

Step 5: interpret the evidence

What the model did well

- Improved accuracy from 0.800 to 0.855.
- Every predicted defect was actually defective.
- Produced no false alarms in the test set.

What the model did poorly

- Detected only 3 of 11 defective runs.
- Missed 8 defective runs.
- Defect recall was only about 27%.

Practical meaning: the model is conservative and should not be used as the sole quality-control mechanism.

Step 6: document limitations and next step

Limitations to report

- Only 220 observations.
- Only four process variables.
- One train/test split.
- No external or prospective validation.
- Associations do not prove causes.

Recommended next step

- Evaluate lower decision thresholds.
- Test class weighting.
- Collect more defective-run examples.
- Examine errors by machine, material, or operating condition.
- Use cross-validation before drawing stronger conclusions.

Rule of thumb: limitations are part of the evidence package

Step 7: completed AI evidence package

Field	Example entry
Problem	Predict whether a process run will produce a defect
Inputs	Temperature, pressure, speed, vibration
Baseline	Majority-class prediction: accuracy 0.800; recall 0.000
Model	Standardized logistic regression
Result	Accuracy 0.855; precision 1.000; recall 0.273; F1 0.429
Boundary	Does not prove process variables cause defects
Next step	Adjust threshold, test class weighting, collect more defect data

Step 8: prepare the evidence for an LLM

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You are assisting a process engineer. Use only the evidence provided below.
Do not invent causes, variables, or performance results.

Evidence:
- Dataset: 220 process runs; 20% defective.
- Inputs: temperature, pressure, speed, vibration.
- Baseline: accuracy 0.800; defect recall 0.000.
- Logistic regression: accuracy 0.855, precision 1.000, recall 0.273, F1 0.429.
- Confusion matrix: TN=44, FP=0, FN=8, TP=3.
- Limitations: small dataset, single split, limited inputs, no external validation.

Produce: technical interpretation, manager summary, next analysis, and one unsupported claim.
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Prompt pattern: role + evidence + constraints + requested structure.

Grounded Response Structure

Section	Content
Technical interpretation	Model improves accuracy but has low defect recall; screening value is limited.
Manager summary	Few false alarms, but only about one-quarter of defects detected. Do not replace inspection.
Recommended next analysis	Evaluate precision–recall tradeoffs across probability thresholds.
Unsupported statement	Do not claim that measured variables cause defects.

The AI explanation should organize evidence — not create new evidence.

Compare AI responses: fluent vs evidence-based

Use the comparison to teach AI auditing.

⚠️ Ungrounded response

May sound confident but can invent causes, prescribe corrective actions, and ignore low recall.

✓ Grounded response

Should cite the supplied metrics, state limitations, avoid causal claims, and recommend a defensible next analysis.

Audit question	What to look for
Evidence	Does every number match the supplied metrics?
Interpretation	Does it distinguish association from causation?
Limitations	Does it mention low recall and lack of external validation?
Action	Is the recommendation consistent with the evidence?

Transition to AI: predictive evidence becomes context

Data → Baseline → Model → Metrics → Interpretation → LLM/RAG workflow → Human review

- The model does not become the full AI system.
- The LLM should explain evidence, not invent evidence.
- The human should decide whether the explanation is acceptable.

Closing message

Evidence-grounded AI is a workflow design problem.

Do not ask:
"Can AI explain this?"

→

Ask instead:
"What evidence can AI use, and what must it not claim?"

→

Then design:
Grounding • limits • review • evaluation